

ASP: An Operational Pre-Mortem Skill for LLM Coding Agents

An Experience Report

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Abstract

LLM coding assistants often produce *locally correct* code that omits failure modes a senior engineer would surface in a pre-mortem — a row-level security policy, an external API's rate limit, replica lag during a migration — and production then breaks in ways that are expensive to debug. We call these missing items *shadow points*. This experience report describes Anticipating Shadow Points (ASP), a 13-phase planning protocol shipped as a Claude Code skill that integrates Klein's (2007) pre-mortem with the multi-agent failure taxonomy of Cemri et al. (2025), checks the plan with a validator that does not share the planner's context, following the self-correction-limit literature (Huang et al., 2024; Tyen et al., 2024), and hands execution to a child subprocess with a file-based contract. We document the design, its deployment as an open-source plugin, and preliminary observations from five engineering evaluations, read by the author without blinding and with transcripts not preserved. A deterministic adjudicator ships with the skill; its scores on author-written fixtures exercise the instrument and nothing more, so we report no measured effect of ASP. We state in advance the result that would show one: in a real-mode benchmark with blind, independently written expected lists and eight replicates per task and condition, a measured benefit if the assisted condition's mean coverage beats the unassisted one on all five tasks and a one-sided, task-stratified permutation test gives $p < 0.05$; no measurable benefit if it does not beat it, ties included, on three or more; inconclusive in between; and no verdict at all if the budget stops collection early. The protocol is drafted and not yet deposited.

Keywords: LLM coding agents; pre-mortem; multi-agent failure modes; Claude Code; operational protocols; experience report

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DOI https://doi.org/10.5281/zenodo.20276631 *(concept DOI of the companion repository · resolves to its latest deposited version; this version of the paper is deposited with release 2.0.0, and until then the DOI resolves to v1.0.3, which this version supersedes)*

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Replication Every number Sections 5.3–5.4 compute from the repository's data and fixtures is re-derived by python run_all.py in the companion repository and asserted by a test; see Appendix B.

Companion repository Flores, C. U. (2026a), https://github.com/ulissesflores/anticipating-shadow-points

Companion whitepaper Flores, C. U. (2026c). *Graceful Refusals as Unsignalled Failures: Characterising claude -p Exit-Code Semantics on Agent Declines.* See paper/graceful-refusals.md.

Resumo

Assistentes de código baseados em LLM muitas vezes produzem código *localmente correto* que omite modos de falha que um engenheiro sênior levantaria num pre-mortem — uma política de segurança por linha, o limite de taxa de uma API externa, o atraso de réplica durante uma migração —, e a produção então quebra de formas caras de depurar. Chamamos esses itens ausentes de *shadow points*. Este relato de experiência descreve o Anticipating Shadow Points (ASP), um protocolo de planejamento em 13 fases distribuído como skill do Claude Code, que integra o pre-mortem de Klein (2007) à taxonomia de falhas multiagente de Cemri et al. (2025), confere o plano com um validador que não compartilha o contexto do planejador, seguindo a literatura sobre os limites da autocorreção (Huang et al., 2024; Tyen et al., 2024), e entrega a execução a um subprocesso filho com contrato em arquivo. Documentamos o desenho, a implantação como plugin de código aberto e observações preliminares de cinco avaliações de engenharia, lidas pelo autor sem cegamento e sem transcrições preservadas. Um adjudicador determinístico acompanha a skill; suas notas sobre fixtures escritos pelo autor exercitam o instrumento e nada mais, por isso não relatamos efeito medido do ASP. Fixamos de antemão o resultado que mostraria um: num benchmark em modo real, com listas esperadas cegas e escritas por revisores independentes e oito réplicas por tarefa e condição, benefício medido se a cobertura média da condição assistida superar a da não assistida nas cinco tarefas e um teste de permutação unilateral, estratificado por tarefa, der $p < 0,05$; nenhum benefício mensurável se não a superar, empate incluído, em três ou mais; inconclusivo nos demais casos; e nenhum veredito se o orçamento interromper a coleta antes. O protocolo está redigido e ainda não foi depositado.

Palavras-chave: agentes de código LLM; pre-mortem; modos de falha multiagente; Claude Code; protocolos operacionais; relato de experiência

1. Introduction

Software teams increasingly delegate routine engineering work — adding a column to a production database, deploying a function that calls an external API, refactoring a utility used across thirty files — to AI coding assistants. When the assistant gets it right, the engineer's day is freed. When it gets it *subtly* wrong — by writing code that compiles, passes its tests, ships, but quietly omits a step a senior engineer would have flagged in advance — production breaks in ways that are expensive to debug. Decision science has a technique for exactly that gap, the pre-mortem (Klein, 2007), and agent research now has a catalogue of how multi-agent LLM systems fail (Cemri et al., 2025). Our thesis is that the two belong together at runtime: before writing code, an agent should be made to run a pre-mortem against a fixed failure taxonomy, and its plan should be checked by a validator that cannot see the planner's reasoning.

The dominant production failure mode of modern LLM coding agents, in our experience, is not *bad* code. It is *incomplete* code: the agent does what was asked, but not what someone with deep domain context would have asked themselves *before starting*. The replica lag during a schema migration. The CORS header on the edge function. The row-level security policy that needs updating to match the new column. We call these missing items **shadow points** — failure modes obvious to a practitioner who has been burnt by them, invisible to an agent that has not.

Two streams of literature converge on this gap. The first, in decision science, is the **pre-mortem**: a structured exercise in which a team imagines a project has already failed and enumerates causes before acting. The technique was operationalized for project teams by Klein (2007) and grounded in the earlier empirical work of Mitchell, Russo and Pennington (1989) on prospective hindsight; Klein (2007) summarises it as raising the ability to identify reasons for a future outcome by about 30%, a figure we could not re-derive from the 1989 paper (Appendix A). The second, in agent

research, is the recent characterization of multi-agent failure modes by Cemri et al. (2025), who derived a 14-mode taxonomy from 150 seed traces of multi-agent LLM systems (inter-annotator agreement κ=0.88) and released a corpus of 1,600+ annotated traces. Neither stream has been integrated as a **runtime discipline** that a coding agent can be made to follow on every non-trivial task.

This paper reports on **Anticipating Shadow Points (ASP)**, a 13-phase planning protocol shipped as a Claude Code skill that combines pre-mortem reasoning with the MAST 14-mode checklist, separates planning from validation under strict prompt isolation, and hands execution to a child subprocess with a file-based contract. We deployed the skill, distributed it through three install paths (plugin marketplace, dev mode, standalone script), ran it through five engineering evaluations, and report the design and what the evaluations can and cannot show. The evaluation is preliminary and, as corrected in this version, unmeasured: GREEN on five tasks and RED on three, one dispatch each, a single model, read by the author without blinding, with the raw transcripts not preserved. We report it as qualitative observation; the four design choices we treat as hypotheses, and a fifth measured in the companion paper, are in Section 6.

1.1. Contributions

This experience report contributes:

1. **The ASP protocol as documented design** — thirteen phases, each emitting a named artifact, organised around the principle that no implementation begins before shadow points have been surfaced.
2. **Twelve operational constraints we call "Iron Laws"**, each tied to a primary literature source or to a documented design decision. Two have a shell-level check in the runner

template the agent is told to run (Iron Law 12's JSON output, a template default the operator can override, and Iron Law 9's ASP_IN_GOAL flag set for the child, whose check at entry is a Bash snippet the skill tells the agent to run); the other ten are prompt-level instructions.

3. **Preliminary, qualitative observations** from five engineering evaluations, and a deterministic adjudicator with cryptographic provenance, exercised on constructed fixtures and ready to score real transcripts without trusting the author's adjudication (the expected lists are still the author's, and the optional synonym sidecar is not provided for any task).

4. **A separate empirical finding** on claude -p exit-code semantics under graceful agent refusal, characterised in the companion paper Flores (2026c). The finding motivated two of the Iron Laws but warrants independent characterization at larger N.

This is an **experience report** in the systems sense, not a benchmark paper. We document a system we built, a methodology we deployed, and what we observed; we do not claim a robust empirical contribution to the LLM-agents benchmark literature. Section 7 enumerates the threats to validity in detail.

2. Background

We review four threads of recent prior work that directly inform ASP's design (§§2.1–2.4), then situate ASP within the older software- and quality-engineering disciplines it transposes (§2.5). We keep the recent-work review short relative to typical agent-systems papers, because the contribution here is operational and the literature context is in service of that.

2.1. Pre-mortem and prospective hindsight

The pre-mortem technique asks a team to imagine, before commitment, that a project has already failed catastrophically, and to enumerate causes. The empirical basis is Mitchell, Russo and Pennington (1989), who showed that imagining a future event as certain (rather than uncertain) yields explanations that are longer and more episodic; Klein (2007) summarises the effect as about a 30% gain in correctly identifying reasons for a future outcome. Klein (2007) operationalized the finding as the project-team "pre-mortem" and the technique has since spread through decision-science and project-management practice. Its experimental base is thin: we found no replication or meta-analysis of the pre-mortem effect in Crossref, arXiv or a general web search (November 2026), which is an absence in those indexes, not proof that none exists. Risk-assessment standards catalogue neighbouring techniques — scenario analysis and the structured what-if technique (SWIFT) — but the table of contents of IEC 31010:2019, Annex B, does not list the pre-mortem by name. The term *prospective hindsight* has also been reused for an unrelated reinforcement-learning signal (J. Zhang et al., 2026); we use it only in Mitchell et al.'s sense. We found no prior integration of the pre-mortem as runtime discipline for coding agents.

2.2. Multi-agent failure modes (MAST)

LLM agents interleave reasoning and acting in a single loop (Yao et al., 2023), and are increasingly composed into multi-agent workflows whose failure modes compound. Cemri et al. (2025) introduce the **Multi-Agent System Failure Taxonomy (MAST)** with **MAST-Data**, a corpus of 1,600+ annotated traces from seven popular multi-agent frameworks such as AG2 (formerly AutoGen; Wu et al., 2023). From rigorous analysis of 150 seed traces (inter-annotator agreement κ=0.88), they derived fourteen failure modes organised into three categories: specification and system design (5 modes), inter-agent misalignment (6 modes), and task verification and

termination (3 modes). The 14-mode list is the strongest recent empirical grounding for a failure-mode checklist; ASP adopts the mode names verbatim and operationalizes each into a pre-mortem prompt plus mitigation in its companion skill files. Two concurrent works extend the taxonomy direction: Zhu et al. (2025) introduce AgentDebug and the AgentErrorTaxonomy (memory / reflection / planning / action / system-level), and S. Zhang et al. (2025), an ICML 2025 Spotlight, opens **automated failure attribution** with the Who&When dataset. MAST is therefore one taxonomy among several: TRAIL formalises a taxonomy of agentic errors over 148 human-annotated traces (Deshpande et al., 2025), AgentRx annotates 170 failed trajectories with a critical step and a category from a grounded-theory taxonomy (Barke et al., 2026), and AgenTracer reports that current reasoning LLMs attribute failures in multi-agent traces with accuracy generally below 10% (G. Zhang et al., 2025). ASP uses MAST as a prompt library for anticipating failures, not as a classifier of observed ones, so the choice of taxonomy bounds what Phase 2 asks about rather than what it can detect.

2.3. Self-correction limits and validator isolation

The strongest single argument for separating the validator from the planner is the convergent finding across three primary works that LLMs cannot reliably critique themselves under shared context. Huang et al. (2024, ICLR) show that intrinsic self-correction without external feedback often degrades performance. Tyen et al. (2024) decompose self-correction into localization and correction and show empirically that LLMs cannot *localize* their own reasoning errors but can correct them given the location externally — *localization is the bottleneck*. Zheng et al. (2023, NeurIPS D&B) document that strong-LLM judges over-favour same-family-model outputs (a "self-enhancement bias") even when objectively judging quality. Kamoi et al. (2024) survey this literature and conclude that reliable self-correction of reasoning errors requires external feedback

or verifiers rather than the model's unaided judgment; intrinsic self-feedback methods such as Self-Refine (Madaan et al., 2023) improve some generation tasks but do not close the localization gap that Tyen et al. (2024) isolate. Together these findings argue that a validator sharing context with a planner inherits the planner's confidence calibration on shared content, including erroneous content the planner is confident about. ASP's design enforces strict prompt isolation between planner and validator (Iron Law 2 below) as the architectural counter-move. Kambhampati et al. (2024) reach the same conclusion for planning: auto-regressive LLMs cannot by themselves plan or self-verify, but can serve as approximate knowledge sources inside an "LLM-Modulo" loop with external verifiers — the role ASP assigns to the isolated validator and to the executable acceptance tests.

2.4. Closest related work: PreFlect

The closest concurrent work is **PreFlect** (Wang et al., 2026), which similarly shifts the self-reflection paradigm from post-hoc correction to pre-execution criticism of agent plans, distilling planning errors from historical agent trajectories and adding a dynamic re-planning mechanism. The high-level similarity is significant: both ASP and PreFlect identify the post-execution-only nature of conventional reflection as a limitation and shift criticism *before* the action. ASP differs from PreFlect along four axes:

1. **Prompt-isolated validator.** PreFlect distils errors into the agent's own pre-execution criticism; ASP enforces a separate-prompt validator subagent that never sees the planner's transcript.
2. **Process-level isolation for the executor.** ASP spawns the executor as a claude -p subprocess with a file-based contract; PreFlect operates within a single process.

3. **Explicit runtime constraints.** ASP states twelve Iron Laws, two of them guarded by executable checks and ten as prompt-level rules; PreFlect is presented as a methodology rather than a constraint system.

4. **An empirical CLI-level finding** characterising claude -p exit semantics under graceful refusal, motivating two of the Iron Laws; this is reported separately in Flores (2026c). The phenomenon itself is documented by the vendor and in public issue reports (catalogued in Flores, 2026c, §2); the contribution there is a measured count and a proposed typed termination contract, not a discovery.

2.5. Classical software and quality-engineering lineage

The four threads above place ASP within the recent LLM-agents literature. A reviewer from software or quality engineering will, however, recognise ASP's machinery as a transposition of disciplines that predate LLMs by decades. We make the borrowing explicit, because concealing it would be both dishonest and weaker: what is new in ASP is not the disciplines but their transposition to a non-deterministic agent runtime and the mechanisms that enforce them there.

Non-redundancy versus defensive re-verification: where ASP departs from Design by Contract. ASP's micro-TODO schema (§3.4) is Design by Contract (Meyer, 1992) relocated from the routine boundary to the agent-task boundary. Meyer's contract assigns obligations across a caller/routine boundary: the caller must establish the PRECONDITION, which the routine is then entitled to *assume* — Meyer's non-redundancy principle forbids the body from re-checking it — and the routine must guarantee the POSTCONDITION on which the caller relies. Of the schema's seven fields, PRECONDITION, ACTION and POSTCONDITION map onto this contract directly, with ACTION as the routine body the pre/post pair wraps. ASP then *departs* from Meyer on

purpose: because the obligor is a non-deterministic model rather than a compiled routine, ASP does not assume the postcondition — it re-verifies it with an external ACCEPTANCE-TEST, and adds a FALSIFICATION-TEST that has no contract-programming analog, importing instead the Popperian stance (Popper, 1959/2002) that a claim earns confidence by surviving a disconfirmation attempt rather than by a confirming check alone. This trade of Design by Contract's non-redundancy for defensive re-verification is the same trust-nothing ethos that produces the exit-code discipline of Iron Laws 11–12; it is a considered inversion of Meyer, not an oversight. The Iron Laws, in turn, are stated as the analogue of DbC's third contract element — the class invariant — rather than as a per-task pre/post condition; ten of them are instructions the agent is told to keep, not checks that run.

Failure-mode enumeration. Shadow-point detection (Phase 2) is a prospective failure-mode enumeration in the tradition of FMEA (MIL-STD-1629A, 1980; IEC 60812:2018). It is deliberately *not* full FMEA: ASP performs no per-component decomposition and no risk-priority scoring (severity × occurrence × detection), so the debt is to FMEA's prospective-enumeration discipline, not its bottom-up mechanics. Three inputs to Phase 2 occupy roles worth separating: the pre-mortem (Klein, 2007; §2.1) supplies the *philosophy* — imagine the failure as already certain; the MAST fourteen-mode taxonomy (Cemri et al., 2025; §2.2) supplies the *library* of failure modes; and the forced enumeration against that fixed library is closer, structurally, to HAZOP-style guideword prompting (IEC 61882:2016) than to component-by-component FMEA. What ASP adds is the substrate — the "component" whose failure modes are enumerated is an LLM plan, and the enumeration is run by a validator that does not share the planner's context (§2.3).

Mistake-proofing. A subset of the Iron Laws (§3.3) are poka-yoke (Shingo, 1986) at the protocol level: the runner template in claude-p-goal-runner.md sets ASP_IN_GOAL for the child and always passes --output-format json (Iron Laws 9 and 12); Iron Law 10's file-based contract is an instruction. These checks live in a Bash template the agent is told to run, not in a hook outside the agent's control, and they are not the physical impossibility of true poka-yoke — an environment flag can be unset or fail to propagate — and the remaining laws are honestly procedural gates (Iron Laws 1, 2, 4, 5) or verification rules that still depend on the operator, of which Iron Law 11 ("never trust $?") is the clearest case: precisely the operator-vigilance dependence poka-yoke is defined against. The borrowing is the mistake-proofing *stance* applied to two shell-level checks in a runner script the agent is instructed to emit, not a claim that ASP eliminates operator vigilance.

Independent verification and validation. ASP's prompt-isolated validator (Iron Law 2) transposes the defining move of Independent Verification and Validation (IV&V; IEEE Std 1012-2016) — a verifier kept independent of the developer, in IEEE 1012's terms technically, managerially and financially — from organisational separation to *prompt* separation: the validator subagent shares no context with the planner it checks. The LLM-specific justification for that isolation is the self-correction literature of §2.3; the *form* of the counter-move is IV&V's.

Dependability vocabulary. The exit-code behaviour behind Iron Laws 11–12 has an exact name in the dependability taxonomy: an exit 0 on a graceful decline is an *unsignalled failure* (Avizienis et al., 2004), neither a crash nor a fail-silent stop. The term, its prior art, and the measurement are developed in the companion paper (Flores, 2026c); this section imports only the vocabulary and does not re-open the novelty question here.

Empirical warrant. The preference for acceptance and falsification tests over trusted status codes has an empirical basis in the adjacent pathology. Yuan et al. (2014) found that a large majority of

catastrophic failures in distributed data-intensive systems trace to faulty handling of non-fatal errors that *were* signalled. ASP's exit-code case is the complement — the signal was *absent* — so the two do not share a thesis; the narrow lesson they do share is that a status channel is not a proxy for outcome, which is what an external ACCEPTANCE-TEST exists to enforce.

Table 1 maps each ASP mechanism to its classical ancestor and states what ASP adds.

Table 1
ASP mechanisms and their classical ancestors

ASP mechanism	**Classical ancestor**	**What ASP adds (and does *not* claim)**
micro-TODO PRE/ACTION/POST + ACCEPTANCE-TEST	Design by Contract (Meyer, 1992)	a non-deterministic obligor whose postcondition is re-verified by external command, not assumed; not the contract idea
micro-TODO FALSIFICATION-TEST	Popperian falsification (Popper, 1959/2002)	a disconfirmation step at the agent-task boundary; no Design-by-Contract analog
Shadow-point detection (Phase 2)	FMEA-style enumeration (MIL-STD-1629A; IEC 60812:2018) + HAZOP guidewords, over the MAST library (Cemri et al., 2025)	enumeration of an LLM plan under isolation; no risk-priority scoring, no per-component decomposition
Iron Laws 9 and 12 as shell-level checks	Poka-yoke (Shingo, 1986)	two checks in a runner script the agent is instructed to emit; not physical impossibility, and Iron Law 10 is an instruction
Prompt-isolated validator (Iron Law 2)	IV&V (IEEE Std 1012-2016)	organisational independence recast as prompt isolation; not IV&V
Never-trust-$? (Iron Laws 11–12)	Unsignalled failure (Avizienis et al., 2004); Yuan et al. (2014)	characterised in the companion paper (Flores, 2026c); not the failure class

Read together, the section makes a deliberately modest claim: ASP's machinery is an *integration* of established reliability and quality-engineering disciplines — contract, failure-mode enumeration, IV&V, mistake-proofing — into one protocol whose contribution is operational rather than conceptual. What that integration adds beyond its ancestors is their application in a non-deterministic agent runtime: the Iron Laws (two of them guarded by executable checks), the prompt- and process-level isolation of §3.2, and the reproducible deterministic adjudicator of §5,

with the claude -p exit-code finding characterised separately in the companion paper. This is the experience-report claim of §1, not a novelty claim against the classical literature.

3. The ASP Protocol

3.1. Overview

ASP is a 13-phase protocol that progresses from intake through delivery sign-off. Each phase emits a named artifact (Table 2). The principle organising the phases is that no implementation begins before shadow points have been surfaced (Phase 2), validated independently (Phase 5), and approved by the user (Phase 7); execution is delegated to a child subprocess with a file-based contract (Phase 9).

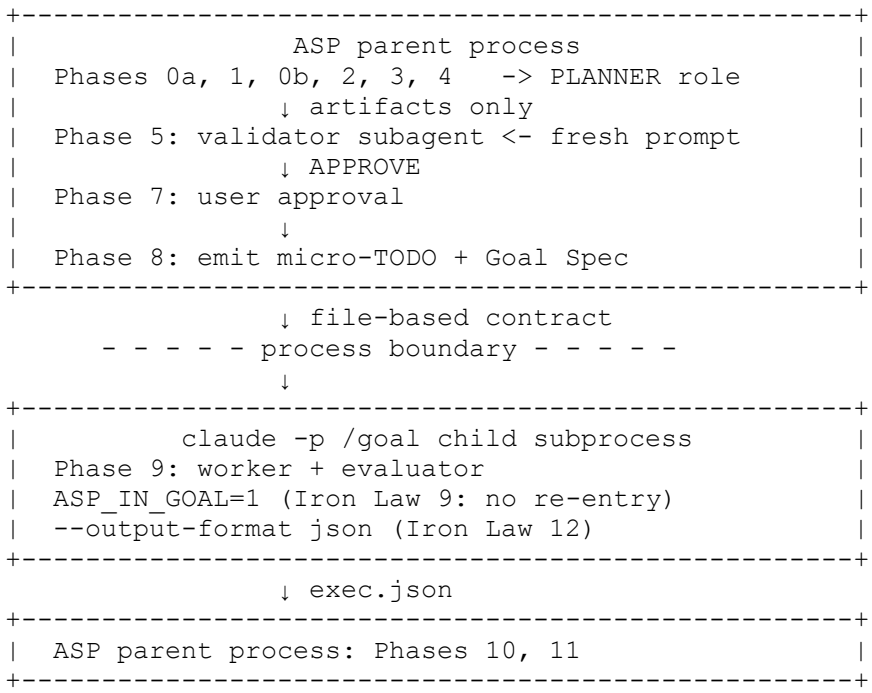
Table 2
ASP phases and their emitted artifacts. Phase 0 has two sub-phases (0a before research, 0b after it), so thirteen phases occupy fourteen rows

#	Phase	Artifact	Design rationale
0a	Pre-research Q&A	intake block	Anthropic Skills best-practice: verifiable intermediate outputs
1	Parallel research	Research summary + recall block	Subagent dispatch pattern
0b	Post-research Q&A	Refined intake (skippable)	Same as 0a
2	Shadow-point detection	Categorised list + mitigations	Klein (2007) + Cemri et al. (2025)
3	Project Charter	Filled charter	Durable artifact-based handoff
4	Macro plan + register	Macro plan + deliverables D01–D*n*	Plan-and-Act (Erdogan et al., 2025)
5	Independent validator	APPROVE / REVISE	Huang et al. (2024) + Tyen et al. (2024) + Zheng et al. (2023)
6	Internal loop (cap=3)	Final-revision plan	Bounded iteration; FM-1.5
7	User approval	Go / no-go	Human-in-the-loop gate
8	Micro-TODO + Goal Spec	TaskCreate per step + Goal Spec	Executor handoff
9	Execution via /goal subprocess	Tasks marked completed with evidence	Process-level isolation; see Flores (2026c)
10	Per-deliverable sign-off	Each deliverable accepted or rejected	Acceptance contract
11	Execution Report + memory	Closure artifact + memory write-back	Reflexion-style memory (Shinn et al., 2023)
12	(Opt-in) public release	Plugin marketplace + GitHub release	Distribution layer

3.2. Architectural positioning

ASP separates three agent roles with explicit artifact-only handoffs (Figure 1):

Figure 1
ASP's three-role separation



Note. Planner phases run in the parent. The validator runs in the parent but in a fresh subagent prompt that receives only artifacts, never the planner's transcript. The worker runs in a child claude -p subprocess with a file-based contract.

3.3. Iron Laws

We state twelve operational constraints and call them Iron Laws (Table 3). Iron Laws 9 and 12 have shell-level checks in the runner template and the entry snippet the agent is told to run; the others are rules the skill instructs the agent to follow. Compliance with either kind has not been measured. Each is mapped to its derivation: a primary literature source, a documented Anthropic recommendation, or an empirical design decision.

Table 3
The 12 Iron Laws and their derivation

#	Law	Derivation
1	No implementation without shadow-point pass first	Klein (2007) + Cemri et al. (2025)
2	No plan approval without independent validator pass (separate prompt)	Huang et al. (2024) + Tyen et al. (2024) + Zheng et al. (2023); mitigates FM-3.3
3	No micro-TODO without macro plan + charter + deliverables approved by user	Anthropic Skills: verifiable intermediate outputs
4	No step completion without fresh acceptance-test evidence	Cemri et al. (2025) FM-3.1 + FM-3.2 + FM-2.6
5	No deliverable accepted without evidence file/output matched	Cemri et al. (2025) FM-3.1
6	No project close without execution report + memory write-back	Shinn et al. (2023) memory buffer
7	No loop without cap (default cap=3 validator rounds)	Cemri et al. (2025) FM-1.5
8	No Phase 9 execution without evaluator dispatch at checkpoints (except under --quick)	Erdogan et al. (2025) planner/executor
9	No reentrant ASP inside claude -p /goal child (ASP_IN_GOAL env-var guard)	Cost-explosion mitigation; design decision
10	No claude -p /goal spawn without file-based contract	Subprocess state-divergence mitigation; design decision
11	Never trust $? from claude -p for semantic success/ failure	Empirical observation, characterised in Flores (2026c)
12	All claude -p invocations must use --output-format json or stream-json	Corollary of Iron Law 11

3.4. The micro-TODO contract

At the boundary between planning and execution (Phase 8), ASP emits a contractual task list. Each task is created via the TaskCreate primitive with seven labeled fields:

- **PRECONDITION** — invariant that must hold before the step.
- **ACTION** — exact command or literal edit.
- **POSTCONDITION** — state change effected by the step.
- **ACCEPTANCE-TEST** — command and expected output that prove the postcondition.
- **FALSIFICATION-TEST** — command or observation that would prove the step failed.
- **DEPENDS-ON** — upstream task identifiers (DAG structure).

- **DELIVERABLE-ID** — which deliverable in the register this step contributes to.

This schema is what makes the task list auditable: no task is marked completed without acceptance-test evidence (Iron Law 4); no deliverable is signed off without matching evidence (Iron Law 5). The schema is shared with the Goal Spec consumed by the Phase 9 subprocess.

4. Operational Deployment

ASP ships as a Claude Code plugin and is distributed through three install paths, each appropriate for a different audience:

1. **Plugin marketplace** (recommended for end users):

   ```
   claude plugin marketplace add ulissesflores/anticipating-shadow-points
   claude plugin install anticipating-shadow-points@anticipating-shadow-points
   ```

   Invocation: /anticipating-shadow-points:asp <task> or bare /asp.

2. **Developer mode** (for contributors): claude --plugin-dir ./ loads the skill in place; useful when iterating on the skill itself.

3. **Standalone install script** (for users who do not use the plugin manager):

   ./scripts/install.sh copies the skill into the user's Claude Code skills directory.

The plugin uses Anthropic's official .claude-plugin/plugin.json manifest format (Claude Code 2.1+; Anthropic, 2026a), and the repository doubles as a self-host marketplace via .claude-plugin/marketplace.json at the repository root.

Permissions. Since plugin version 1.0.4 the Phase 9 runner spawns the claude -p child with **no permission flag**: the child runs under the user's own Claude Code permission settings, and the skill never grants, widens or bypasses them. Earlier versions passed --permission-mode bypassPermissions; that was removed after the Claude plugin directory declined the listing because

a plugin must not grant itself permissions. In non-interactive mode an action the user has not pre-allowed is denied rather than prompted; the parent reads the permission_denials field of the child's JSON result and reports any denial to the user instead of retrying with wider permission. This makes autonomous execution strictly weaker than in the May 2026 deployment, and safer: the skill can only do what the operator already allowed. The skill itself is implemented as a Markdown-based skill following the Anthropic Agent Skills best-practices document (Anthropic, 2026b), with companion files for the MAST checklist, per-mode pre-mortem prompts, and the seven artifact templates referenced in Section 3.1.

Documentation is published as a GitHub Pages site, with translations of the README into Portuguese, Spanish, Italian, and Hebrew, all machine- or AI-assisted: the Portuguese awaits the author's review, the other three a native speaker's (docs/translations.md). The full source — skill files, plugin manifests, documentation, test artefacts, benchmark code — is MIT-licensed; the paper text, figures and data are CC BY 4.0.

5. Preliminary Observations

We ran ASP through five structured engineering evaluations spanning heterogeneous domains. This section separates three things that version 1.0 blurred: what the author observed in real model dispatches (5.2), what the deterministic adjudicator shows about itself on constructed fixtures (5.3), and what has not been measured at all.

5.1. The evaluation set

Table 4
The five engineering evaluations

ID	Domain	Task (abbreviated)	Expected size	Acceptance
E1	Schema migration	NOT NULL column added to 1M-row user_profiles	10	≥ 80%
E2	Edge deploy	Edge function calling rate-limited external API	10	≥ 80%
E3	Refactor	Date-formatting utility used across 30+ files	10	≥ 80%
E4	RLS policy	Row-level security policy referencing new column	10	≥ 80%
E5	Scheduling	Cron job whose script conflicts with disabled service	10	≥ 80% and points 1 and 8 present

Each evaluation specifies an INPUT (a task statement as a senior engineer might phrase it to a colleague), an EXPECTED list of shadow points that a competent practitioner would surface, and an ACCEPTANCE threshold (default 80%). E5 is not portable: two of its ten expected points (1 and 8) depend on the author's private project memory and cannot surface from its INPUT alone, so it measures memory recall as much as planning. The full text of each evaluation is in the companion repository.

5.2. What was observed in real dispatches (author, non-blind)

On 2026-05-17 the evaluations were dispatched to fresh subagents: GREEN (with the ASP discipline injected: pre-mortem framing, MAST 14-mode forced enumeration, mitigation requirement) on all five, and RED (asked to plan as a senior engineer would, without any framework) on three, E1–E3; E4 and E5 were added afterwards and run in GREEN only. All ran on the session model of the time (Claude Opus 4.7; the model was not recorded per dispatch), one dispatch per condition. The subagents inherited the author's global agent instructions, so the RED condition was a competent baseline, not a naive one. The author read each plan against the

expected list and recorded a coverage judgement; the summaries, with the subagent identifiers, are committed in tests/baseline-pressure-tests.md and tests/evals-summary.md. The raw transcripts were not committed and are no longer available. The RED prompts recorded there are shorter Portuguese phrasings of E1–E3 that omit details the expected lists later score: E2's RED prompt does not state Resend's 100 req/s rate limit, and E1's does not say that app code already references the new column in five query sites. Part of the RED shortfall may therefore be an artefact of the prompt, not of the absence of the protocol.

Under that non-blind reading, the three RED plans covered roughly 30–60% of their expected lists (mean near 47%, counting partial mentions as covered) and the GREEN plans covered all or nearly all of theirs. The committed summaries, written in May 2026, conclude that ASP is "reproducibly better"; a note added above them in version 2.0 withdraws that conclusion, which is not supported. These figures are an author's judgement on unpreserved text, scored against lists the same author wrote, with the MAST vocabulary that the GREEN prompt injects appearing in those lists. They are reported as observation, not measurement, and no number in this paragraph is asserted by the replication layer.

5.3. What the deterministic adjudicator shows about itself

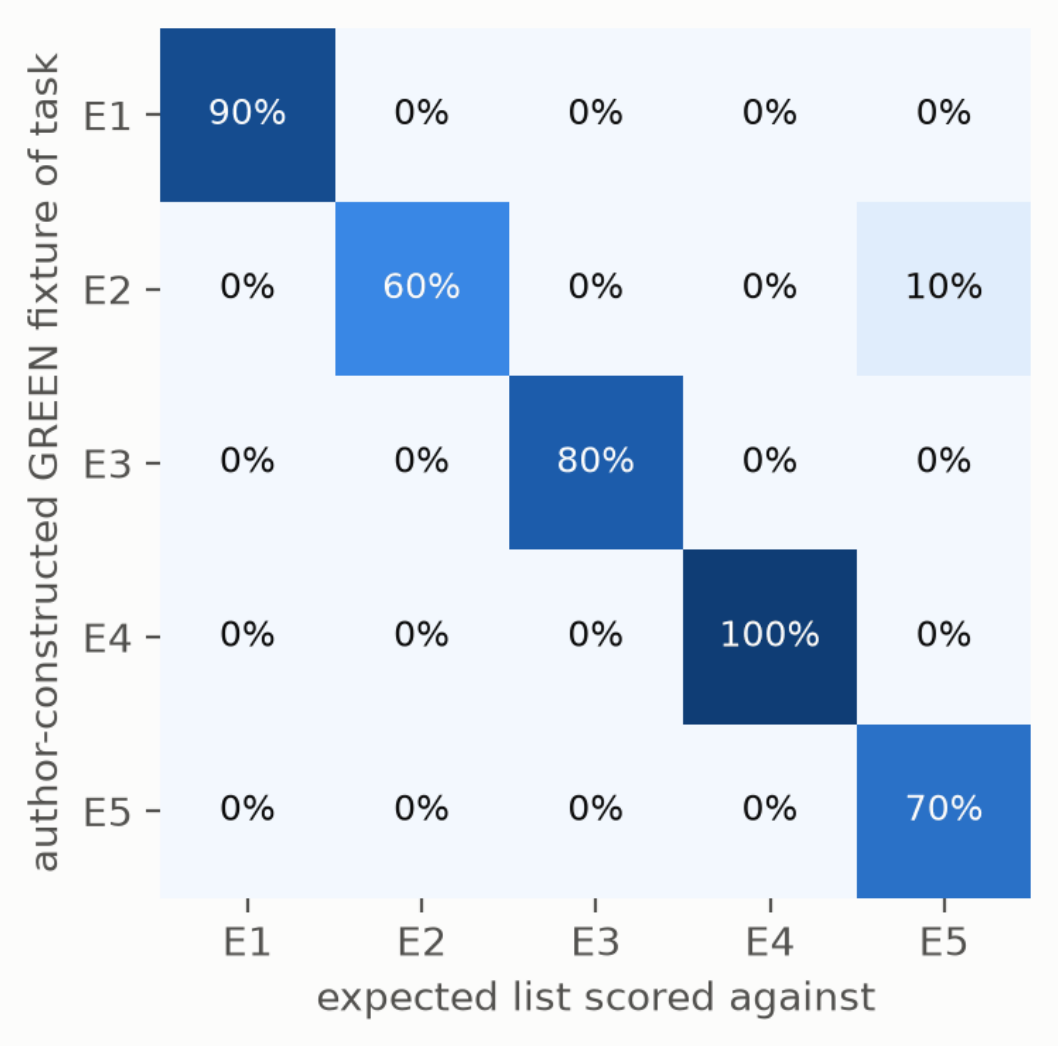
The benchmark/ package scores a plan against an expected list with a deterministic 4-layer matcher (exact name, synonym sidecar, Jaccard sliding window, unmatched). On the published run every one of the 42 matches comes from the first layer (exact name); no synonym sidecar exists for any task, and the Jaccard layer never fires. Its one published run is a **mock-mode** run: the RED and GREEN texts are fixtures the author wrote to exercise the pipeline end to end (benchmark/tests/fixtures/mock_runs/); on them layers 2 and 3 never produce a match (layer 3's overlaps are computed and stay below the threshold), and only the unit tests exercise a match

through those layers. The fixtures themselves were written deliberately weaker for RED and deliberately aligned with the expected list for GREEN. On those fixtures the matcher scores RED 4% and GREEN 80% on average — a statement about how the fixtures were written, not about ASP.

The useful property the mock run does establish is **discrimination**: scoring each GREEN fixture against a *different* task's expected list drops the mean from 80% to 0.5% (largest off-task cell 10%; Figure 2). That is weaker evidence than it looks: the one off-task hit is the generic one-word name "Alerting", and the other cells are zero because layer 1 requires a whole expected name to appear verbatim. The matcher is also stricter than the evaluation files, which tell a human evaluator to accept paraphrases. Recall is a separate matter, and the fixtures show a false negative: the E2 RED fixture mentions a rate limit, retries and logging and still scores 0/10: its best window overlap with the expected point "Rate-limit backoff" (its name and description) is a Jaccard of 0.11, below the 0.30 threshold, and the matcher does no stemming ("logging" does not match "logs"). Paraphrase with low lexical overlap is missed. Re-scoring also exposed a defect, fixed in v2.0: the sliding window skipped the last few tokens of a text whose length did not land on the stride; re-scoring every fixture with the fix changes no coverage value, though the best overlap of three unmatched points rises slightly (E2 RED "Observability / logs" from 0.00 to 0.02), still far below the threshold. Layer 3 also has an arithmetic ceiling: Jaccard divides by the union, and a 40-token window of the fixtures holds 27 to 40 distinct content words (median 36); in any window with 34 or more, an expected point with ten or fewer distinct tokens cannot reach 0.30 even if the plan repeats all of them, and 19 of the 50 expected points are that short. The real-mode protocol therefore changes layer 3 before deposit (PREREGISTRATION.md). The matcher has not yet been applied to any real transcript. The real-mode protocol, with expected lists re-authored blind

by independent engineers, is specified in the repository's PREREGISTRATION.md and has not been run. Its decision rule is fixed in advance: with eight replicates per task and condition (80 dispatches), a run is labelled incomplete, with no verdict, if the budget stops collection before every task reaches eight replicates per condition; otherwise a benefit is reported only if the assisted condition's mean coverage beats the unassisted one on all five tasks and a one-sided, task-stratified permutation test on the mean per-task difference (10 000 relabellings, seed asp-bench-v1) gives $p < 0.05$; no measurable benefit if the assisted condition does not beat the unassisted one (GREEN − RED ≤ 0, so a tie counts) on three or more tasks; inconclusive in between. Coverage will be reported per matcher layer, since layer 1 rewards a plan that reuses the expected names verbatim.

Figure 2
Instrument check: each author-constructed GREEN fixture scored against every task's expected list. Not an effect of ASP



Which component would matter most, if ASP has an effect, is discussed as a hypothesis in Section 6.

5.4. The claude -p exit-code finding

During a four-test battery against the claude -p non-interactive runner, we observed that the binary returns exit code 0 even when the agent gracefully refuses the requested goal — the harness reports a successful conversation end despite the agent semantically refusing the task. The exit code

reflects process-level success of the CLI binary, not semantic success of the agent's task. Iron Laws 11 and 12 require --output-format json or stream-json and reading is_error, terminal_reason and stop_reason. Those fields separate a turn-cap cut-off and a classifier refusal from a completed turn, but not a model-generated decline from a delivery; the runner therefore also flags replies whose text matches a decline pattern, a heuristic the companion paper tests.

This finding is observationally significant for any toolchain that wraps claude -p in a shell pipeline. Because the observation was from a single refusal scenario in our own battery, we **characterise it independently** in the companion paper Flores (2026c), which reports an N=50 measurement across four categories of refusal-provoking prompt (impossible-in-one-turn, capability, safety, ambiguity) on Claude Code 2.1.200 in July 2026. The session model was Fable 5; Opus 4.8 answered 16 of the 50 prompts, including all 12 safety prompts — consistent with an automatic fallback hand-off, although the envelopes record which model answered, not why. Twenty-four trials exited 0, all with stop_reason: end_turn. The companion paper counts the declines among them four ways: 20 (40%, 95% Wilson CI 28–54%) under the author's non-blind label; 19 (38%) under the majority of a codebook, written after the author had read the replies, and applied by five LLM raters, four of which returned identical labels, all of the lineage of the model under test, so the majority is a codebook check, not an inter-rater agreement estimate; 12 (24%) counting only the safety and destructive-command refusals and excluding the eight reasoned declines of impossible-in-one-turn tasks; and 9 (18%) under the locked English-only phrase detector, which misses the replies written in Portuguese. An earlier snapshot (May 2026, Claude Code 2.1.143, Opus 4.7) read 60% under the same detector, on a different model and reply language. A human second rater and a third snapshot under a drafted pre-registration are the open follow-ups.

6. Design Hypotheses

We highlight five design choices we believe are load-bearing. The first four are hypotheses about ASP that no data here measures, each stated with the observation that motivated it; the fifth rests on the companion paper's measurement.

Hypothesis: a forced taxonomy surfaces more than free-form pre-mortem. In the author's drafts, free-form pre-mortem clustered on obvious risk categories (data, integration) and underweighted less-salient ones (time/timezone, observability, operational excellence, contract drift). No dispatch tested free-form pre-mortem as a condition (RED had no framework at all), so this rests on the author's reading of drafts. The free-form prompt ("imagine the task has failed in 30 days; list 10+ causes") would then be necessary but not sufficient, and the MAST 14-mode interrogation would surface categories the free-form pass clusters away from.

Hypothesis: prompt isolation of the validator matters. We expect validators that share context with planners to reproduce the planner's confidence on shared content. The convergent literature on intrinsic self-correction limits, error-localization failure, and judge self-enhancement bias all point at this; we have not tested it inside ASP.

Hypothesis: file-based handoffs survive context pressure better than transcript-based ones. ASP commits charter, macro plan, deliverables register and goal spec to disk before the executor is spawned. The child receives only the Goal Spec text on stdin and returns a JSON result; the parent reconstructs state from that result and from the files on disk, so neither side depends on a transcript that may have been compressed or truncated. We have not compared the two handoffs.

Hypothesis: stated constraints (Iron Laws) make a skill harder to skip. In our experience, skills that read as "guidelines" are often skipped under context pressure. Skills that emit a checklist

of non-negotiables, with templates whose fields must be filled before the next phase, are in our use harder to skip. Compliance has not been measured.

Measured elsewhere: treating claude -p exit codes as semantic-success signals is a silent footgun. This is the finding reported in Flores (2026c); mentioning it here because the design implication (Iron Laws 11 and 12) was material to the system's reliability.

7. Threats to Validity

We treat threats explicitly across the four categories standard in empirical software-engineering reports.

Construct validity. The same researcher who designed the protocol authored the expected-shadow-points lists *and* hand-adjudicated coverage. Even with a written rubric, this is the single largest threat to the observation's interpretability. The deterministic adjudicator we ship would mitigate this once applied to real, preserved transcripts; in this version it has only been run on constructed fixtures (§5.3), and it does not address the prior question of whether the expected lists themselves are biased toward what ASP surfaces. **Blind re-authoring by independent reviewers is the strongest follow-up.**

Internal validity. With one dispatch per cell, GREEN on five tasks and RED on three, the evaluation at most suggests that the protocol can change the transcripts on these cases; it is *not* sufficient to estimate an effect size, establish causality, or support statistical inference, and no statistical test is reported. No formal ablation of components has been run (the qualitative attribution of load-bearing-ness to the MAST checklist in §6 is a hypothesis). The acceptance threshold of 80% is the author's choice and not derived from prior literature.

External validity. Single underlying model (Claude Opus 4.7, presumed from the session; not recorded per dispatch), single environment (the author's macOS workstation and the Claude Code version of May 2026, not recorded per dispatch), five engineering tasks drawn from the author's domain experience. We make no claim of generalisation beyond these conditions.

Reliability. All evaluation INPUT statements, expected-shadow-point lists, the mock-run fixtures and the deterministic-adjudicator code are committed to the companion repository; the 2026-05-17 dispatch transcripts were not preserved, so the observations of §5.2 cannot be re-checked. The benchmark package builds a cryptographically-chained manifest per run so any byte edited under a run directory invalidates the manifest hash.

8. Limitations and Future Work

The limitations of this experience report match the threats above one-to-one. The highest-priority follow-ups, in order, are:

1. **Blind re-authoring** of the expected-shadow-point lists by 2+ independent reviewers with no exposure to ASP, then a real-mode run whose transcripts are preserved and hash-sealed, adjudicated blind. Compute Cohen's κ for inter-adjudicator agreement; re-measure coverage.
2. **Formal ablation** of the load-bearing hypothesis: MAST checklist on/off (testing the hypothesis of §6), validator prompt isolation on/off, Phase 0a intake on/off. No ablation protocol exists yet; Protocol 2 of PREREGISTRATION.md covers only RED vs GREEN, at 8 replicates per cell with a fixed budget rather than a power analysis.
3. **A human second rater and a third claude -p snapshot** for the companion finding (Flores, 2026c, §8), the latter under Protocol 1 of PREREGISTRATION.md.

4. **Cross-model evaluation.** Re-run on Claude Sonnet, GPT-5, Gemini 2.5, and an open-weight model.
5. **Head-to-head against PreFlect.** Run ASP and a PreFlect reimplementation on the same task set.
6. **Embedding-based adjudicator** as a v0.2 of the benchmark, with pinned sentence-transformer model, to raise the recall floor on lexically-disjoint paraphrases.
7. **Standardized-benchmark replication.** The five-task set is not a public agent benchmark; replicating the direction on SWE-bench (Jimenez et al., 2024) or a comparable standardized suite would test whether it survives outside the author's chosen tasks.
8. **Translation review**: the author's review of the Portuguese and a native speaker's of the Spanish, Italian and Hebrew documentation translations.
9. **Submission target.** Workshop venues — NeurIPS LLM-agents track, ICLR agents workshop, AAAI agents track — are the appropriate fit for the current empirical scale. Main-track submission requires the follow-ups above.

9. Conclusion

We described ASP, an operational pre-mortem skill for LLM coding agents, integrating Mitchell, Russo and Pennington's (1989) prospective hindsight finding, Klein's (2007) pre-mortem operationalisation, and Cemri et al.'s (2025) MAST 14-mode failure taxonomy as runtime discipline under a contract of twelve Iron Laws. The skill ships open source with three install paths. On the three tasks run in both conditions the author observed, without blinding and on transcripts that were not preserved, that the assisted plans covered more of the expected failure modes; no

effect has been measured. The deterministic adjudicator shipped with the skill has been exercised only on constructed fixtures, where it separates whole expected names but has no measured recall; a real-mode run with independently authored lists, drafted for pre-registration, is the decisive next step.

We additionally observed that claude -p returns exit code 0 even when the agent gracefully refuses the requested goal. The vendor documents the behaviour for its API; the companion paper (Flores, 2026c) adds a count at the CLI boundary.

The strongest follow-up is blind re-adjudication. The most useful companion work is a human second rater and a pre-registered third snapshot of the claude -p measurement (Flores, 2026c).

Acknowledgments

This experience report was prepared in explicit human-in-the-loop collaboration with Claude (Anthropic, Opus 4.7), used as a drafting, verification, and review partner. Methodology decisions, empirical design, and finalisation were performed by the human author. The collaboration model is documented in the companion repository's docs/ARCHITECTURE.md Parts 5–6.

The author thanks the maintainers of the obra/superpowers skill framework, the contributors of the open agentskills.io standard, and the Anthropic Claude Code team for the /goal primitive whose empirical behavior motivated Iron Laws 11 and 12.

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Appendix A — Citation verification status

Every claim in the body cites a primary source where one exists. The paragraphs below record when each group of references was checked.

The references present in version 1.0 were verified by direct URL fetch on 2026-05-17; references added in the July 2026 revision (Flores, 2026c; the classical-lineage entries of Section 2.5) were checked when added; and the entries added in November 2026 (Barke et al., 2026; Deshpande et al., 2025; Kambhampati et al., 2024; G. Zhang et al., 2025; J. Zhang et al., 2026) were checked against their arXiv records, and Cemri et al. (2025) against its NeurIPS proceedings record. The MAST 14-mode names are quoted verbatim from Cemri et al. (2025) and listed in the companion skill file skills/anticipating-shadow-points/mast-checklist.md. Mitchell et al. (1989) is the primary study of prospective hindsight; the ≈30% figure is Klein's (2007) summary of it, which we could not re-derive from the 1989 paper, and Klein (2007) is also the operational recipe.

The nine references introduced in §2.5 (Avizienis et al., 2004; International Electrotechnical Commission [IEC 60812:2018], 2018; International Electrotechnical Commission [IEC 61882:2016], 2016, checked against the IEC catalogue on 2026-11-07; IEEE [Std 1012-2016], 2017; Meyer, 1992; Popper, 1959/2002; Shingo, 1986; U.S. Department of Defense [MIL-STD-1629A], 1980; Yuan et al., 2014) were verified against their primary source — DOI, ISBN, or standard catalogue entry — on 2026-07-08.

Appendix B — Evaluation set and reproducibility

The five engineering evaluations are committed in the companion repository:

- skills/anticipating-shadow-points/evals/01-supabase-migration.md (E1)
- skills/anticipating-shadow-points/evals/02-edge-function-deploy.md (E2)
- skills/anticipating-shadow-points/evals/03-refactor-shared-util.md (E3)
- skills/anticipating-shadow-points/evals/04-rls-policy-change.md (E4)
- skills/anticipating-shadow-points/evals/05-cron-skill-conflict.md (E5)

Each file contains the INPUT statement, the EXPECTED shadow-points list, and the ACCEPTANCE threshold, all authored by the protocol's designer. The benchmark package in benchmark/ provides:

- a deterministic 4-layer adjudicator (exact / synonym / Jaccard / unmatched);
- per-run cryptographic provenance (env, code, inputs, outputs and adjudication hashed into a Merkle-style chain);
- a verify subcommand that re-walks a run directory and reports any byte-level drift;
- 31 unit tests covering the adjudicator, the evaluation and reply parsers, the manifest and the hasher, including the determinism contract.

Reproducing the computed numbers of Sections 5.3–5.4 (no model calls). From a clone of the repository, python run_all.py re-scores the author-constructed fixtures of the mock benchmark run with the shipped adjudicator, recomputes the discrimination matrix, regenerates Figure 2, asserts each number (tests/test_paper_numbers.py, which also asserts that the run is labelled mock) and verifies a single SHA-256 chain hash over code, tests, frozen inputs, data and

results (output/hash-chain.md). The author's observations in Section 5.2 cannot be reproduced: their transcripts were not preserved. This version supersedes v1.0.3 (https://doi.org/10.5281/zenodo.20277015), which presented the adjudicator's scores on these fixtures as a reading of model transcripts.

Collecting real evidence (costs model calls). Install the benchmark package and run asp-benchmark run --mode real --model <model> followed by asp-benchmark verify <run-id>. This has not been done for any published number, and today's command is not yet Protocol 2: it sends a single-turn template rather than the shipped skill, runs one replicate per task in a fixed order and passes no clean-room flags. PREREGISTRATION.md lists the changes the runner and the matcher need before any dispatch. The protocol is specified independently of the Python implementation in benchmark/PROTOCOL.md so it can be re-implemented in another language.

About the Author

Carlos Ulisses Flores is CTO and Chief Researcher at Codex Hash Research Laboratory, an independent R&D lab in São Paulo, Brazil. He builds production systems where silent failure is expensive — historically in algorithmic trading, embedded IoT and edge cryptography, and currently in agentic-AI tooling. The ASP skill described in this whitepaper grew out of that latter line of work; the empirical finding reported in the companion whitepaper grew out of using Anthropic's claude -p runner as an autonomous execution kernel and watching it return exit code 0 on a graceful agent refusal.

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